# Daily Algebra Drill #1 — Speed Sheet

*Sheet by David Akinyele-Aje*

| Section | Topic | Questions | Target time |
|---|---|---|---|
| A | Rapid Recognition | 10 | 2 min 30 sec |
| B | Manipulation Drills | 10 | 8 min |
| C | Substitution & Structure | 8 | 10 min |
| D | Challenge | 5 | 15 min |

*New toolkit today: Difference / Sum of Squares & Cubes · Newton's Identity Chain · Disguised Quadratics · Telescoping & Partial Fractions.*

## Section A Rapid Recognition ($\leq$15 s each)

*Factorise, simplify, or evaluate instantly. No expansion. Pure recognition.*

**1.** Factorise completely: $x^4 - 16$.

Ans: $(x^2+4)(x+2)(x-2)$

**2.** Simplify: $\dfrac{a^2 - b^2}{a^2 + 2ab + b^2}$.

Ans: $\dfrac{a-b}{a+b}$

**3.** Without expanding, evaluate: $103^2 - 97^2$.

Ans: 1200

**4.** Factorise: $2x^2 + 7x - 15$.

Ans: $(2x - 3)(x + 5)$

**5.** Simplify: $\left(x + \frac{1}{x}\right)^2 - \left(x - \frac{1}{x}\right)^2$.

Ans: $4$

**6.** Factorise: $x^3 + 8$.

Ans: $(x + 2)(x^2 - 2x + 4)$

**7.** Simplify: $\dfrac{x^2 - 5x + 6}{x^2 - 4}$.

Ans: $\dfrac{x - 3}{x + 2}$

**8.** If $a + b = 5$ and $ab = 3$, find $a^2 + b^2$.

Ans: 19

**9.** Factorise: $4x^2 - 12x + 9$.

Ans: $(2x - 3)^2$

**10.** Without a calculator, simplify: $\dfrac{1}{1 \cdot 2} + \dfrac{1}{2 \cdot 3} + \dfrac{1}{3 \cdot 4} + \cdots + \dfrac{1}{9 \cdot 10}$.

Ans: $\frac{9}{10}$

## Section B Manipulation Drills ($\leq$50 s each)

*Algebraic fractions, rearranging, hidden structure. Resist the urge to expand.*

**1.** Simplify: $\dfrac{x^3-1}{x^2-1}$.

Ans: $\dfrac{x^2+x+1}{x+1}$

**2.** Simplify: $\dfrac{1}{x-1}-\dfrac{2}{x^2-1}$.

Ans: $\dfrac{1}{x+1}$

**3.** Make $x$ the subject: $\dfrac{x+a}{x-a}=k \quad (k\neq 1)$.

Ans: $x=a\dfrac{k+1}{k-1}$

**4.** Given that $a+b+c=0$, express $(a+b)(b+c)(c+a)$ in terms of $abc$.

Ans: $-abc$

**5.** Show that $\dfrac{x^2+3x+2}{x^2-x-2}$ simplifies, and state all values of $x$ for which the original expression is undefined.

Ans: $\dfrac{x+2}{x-2}$ ; undefined at $x=2$ and $x=-1$.

**6.** Simplify: $\left(\dfrac{x^2}{y}-\dfrac{y^2}{x}\right)\div\left(\dfrac{x}{y}+1+\dfrac{y}{x}\right)$.

Ans: $x-y$

**7.** Factorise: $a^2(b-c)+b^2(c-a)+c^2(a-b)$.

Ans: $-(a-b)(b-c)(c-a)$ or $(a-b)(c-a)(c-b)$

**8.** Solve for $x$: $\dfrac{3}{x+1}+\dfrac{4}{x+2}=\dfrac{11}{(x+1)(x+2)}$.

Ans: $x=\frac{1}{7}$

**9.** Simplify: $\dfrac{2^{n+2}+2^n}{2^{n+1}+2^{n-1}}$.

Ans: 2

**10.** Given that $p+q=7$ and $p^3+q^3=133$, find $pq$.

Ans: $pq = 10$

## Section C Substitution & Structure (≤90 s each)

*Look for symmetry, disguised quadratics, and useful substitutions before computing anything.*

**1.** Solve: $x^4 - 5x^2 + 4 = 0$.

Ans: $x = \pm 1, \pm 2$

**2.** If $x + \dfrac{1}{x} = 3$, find $x^3 + \dfrac{1}{x^3}$.

Ans: 18

**3.** Find all real solutions to: $(x^2 - 3x)^2 - 2(x^2 - 3x) - 8 = 0$.

Ans: $x = -1,\ 1,\ 2,\ 4$

**4.** Simplify: $\dfrac{x^2 + y^2}{xy} - \dfrac{(x-y)^2}{xy}$.

Ans: 2

**5.** The expression $x^4 + 4y^4$ can be factorised over the integers. Do so.

Ans: $(x^2 + 2y^2 + 2xy)(x^2 + 2y^2 - 2xy)$

**6.** Solve: $\sqrt{2x+1}+\sqrt{2x-1}=t$ for $x$ in terms of $t$ ($t>0$).

Ans: $x=\frac{t^4+4}{8t^2}$

**7.** Given that $a+b+c=0$, evaluate $\dfrac{a^2}{bc}+\dfrac{b^2}{ca}+\dfrac{c^2}{ab}$.

Ans: 3

**8.** Solve the system: $x+y=5,\quad x^2+y^2=17$.

Ans: $(x,y)=(1,4)$ or $(4,1)$

## Section D Challenge (TMUA / SMC difficulty)

*Insight over computation. Each question has a short, elegant solution — find it.*

**1.** Find the value of

$$\frac{(2025)^3 - (2024)^3 - (2023)^3 + (2022)^3}{(2025)^2 - (2022)^2}.$$

Ans: 2

**2.** Let $f(x) = \dfrac{x}{x+1}$. Define $f^n$ as $f$ composed with itself $n$ times. Find a closed form for $f^n(x)$ and hence compute $f^{2025}(1)$.

Ans: $f^n(x) = \dfrac{x}{nx+1}$ ; $f^{2025}(1) = \dfrac{1}{2026}$

**3.** Positive reals $a, b, c$ satisfy $abc = 1$. Simplify:

$$\frac{1}{1+a+ab} + \frac{1}{1+b+bc} + \frac{1}{1+c+ca}.$$

Ans: 1

**4.** Without solving for $x$, find the value of $x^2 + \dfrac{1}{x^2}$ if $x - \dfrac{1}{x} = \sqrt{5}$. Hence find the exact value of $x^4 + \dfrac{1}{x^4}$.

Ans: $x^2 + \frac{1}{x^2} = 7$; $x^4 + \frac{1}{x^4} = 47$

**5.** Find all integers $n$ such that $n^2 + 4n + 3$ is a perfect square.

Ans: $n = -1, -3$